NOTE The document identifier and heading has been changed on this page to reflect that this is a performance specification There are no other changes to this document The document identifier on subsequent pages has not been changed but will be changed the next time this document is revised

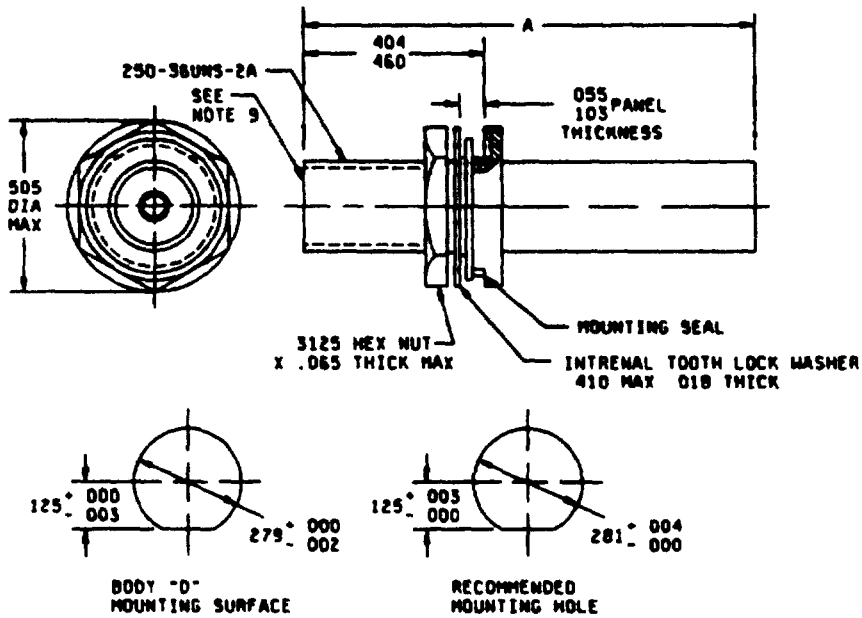
INCH-POUND
MIL-PRF-39012/59F
27 January 1992
SUPERSEDING
MIL-C-39012/59E
3 October 1986

PERFORMANCE SPECIFICATION

CONNECTORS, RECEPTACLE, ELECTRICAL, COAXIAL, RADIO FREQUENCY, (SERIES SMA (CABLED) - SOCKET CONTACT, JAM NUT MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-PRF-39012



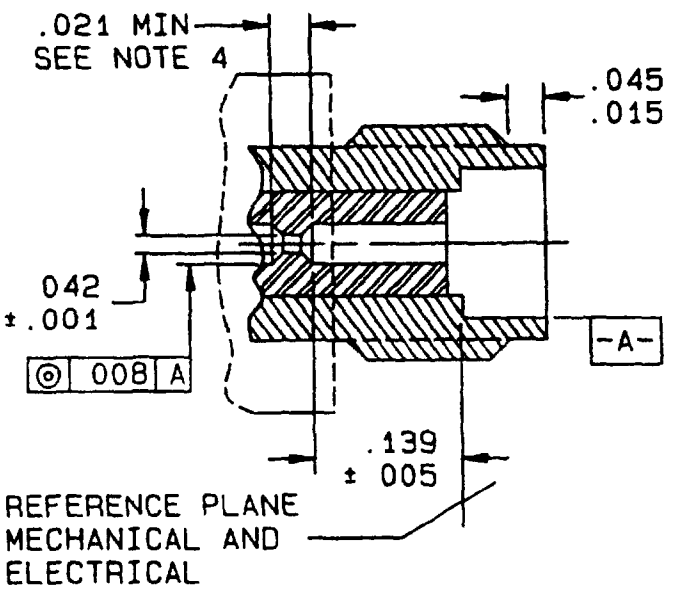
Inches	mm
002	0 05
003	0 08
004	0 10
018	0 46
055	1 40
065	1.65
103	2 62
125	3 18
250	6 35
279	7 09
281	7 14
3125	7 94
404	10 41
410	11 68
505	12 83

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. For dimension A, see tables I and V
4. Dimension 505 (12 83) is the largest overall diameter of the connector
5. Wrench flats are to accommodate standard wrench openings in accordance with FED-STD-H28
6. Dimension A defines the overall length of connector when assembled to the appropriate cable
7. All undimensioned pictoral configurations are for reference purposes only
8. Unless otherwise specified tolerance is ± 005 (0 13)
9. Series SMA, socket contact interface in accordance with MIL-STD-348

FIGURE 1 General configuration

DISTRIBUTION STATEMENT A Approved for public release distribution is unlimited

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CATEGORY D

Inches	mm
.001	0.03
.005	0.13
.008	0.20
.015	0.38
.021	0 53
.042	1 07
.045	1 14
.139	3 53

NOTES
1. Dimensions are in inches
2. Slitting of inner contact optional.
3. Metric equivalents are given for information only.
4. Chamfer is optional, if chamfer is used put chamfer on a 30° maximum

FIGURE 2 Category D captivation detail.

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TABLE I Dash numbers, cross reference and dimensions

Dash number	Applicable cable *	Dimensions	Inches-millimeters maximum 1/
Category A - Field serviceable (no special tools required) 2/			
3006 3/ 4006 3/	M17/93-RG178* M17/169-00001φ	A	960(24 38)
3007 3/ 4007 3/	M17/119-RG174 M17/173-00001φ M17/113-RG316* M17/172-00001φ		
3008 3/ 3008 3/	M17/54-RG122* M17/157-00001φ		
3009 3/ 4009 3/	M17/28-RG058 M17/60-RG142@ M17/84-RG223* M17/155-00001φ M17/158-00001φ M17/167-00001φ		
3010 3/ 4010 3/	M17/111-RG303*@ M17/170-00001φ		
3030 3/ 4030 3/	M17/152-00001		
Category C - Field replaceable (MIL-C-22520/5-01 basic crimp tool) 4/			
3025 3/ 4025 3/	M17/93-RG178** M17/169-00001**φ	A	1.125(28.58)
3026 3/ 4026 3/	M17/119-RG174Σ M17/173-00001Σφ M17/113-RG316Σ* M17/172-00001Σφ		
3027 3/ 4027 3/	M17/54-RG122*& M17/157-00001φ&		

See footnotes at end of table

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TABLE I Dash numbers, cross reference and dimensions - Continued

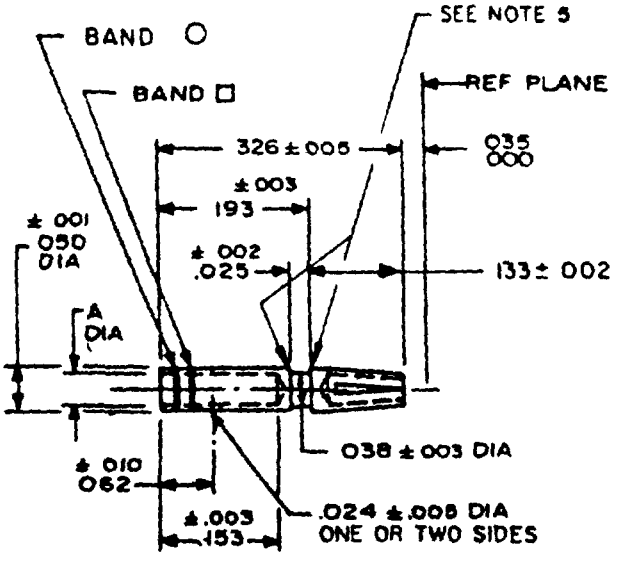
Dash number	Applicable cable *	Dimensions	Inches-millimeters maximum 1/
Category C - Field replaceable (MIL-C-22520/5-01 basic crimp tool) 4/			
3028 3/ 4028 3/	M17/60-RG142*a- M17/158-00001-ϕ M17/167-00001-ϕ M17/84-RG223-	A	1 125(28 58)
3029 3/ 4029 3/	M17/111-RG303- M17/155-00001-ϕ M17/28-RG058*- M17/170-00001-ϕ		
Category D - Field replaceable - Defined piece parts 4/ 5/ 6/			
3502 3/ 4502 3/	M17/60-RG142a* M17/158-00001ϕ M17/128-RG400 M17/175-00001ϕ	A	1 265(32 13)

1/ Millimeters are in parentheses
2/ All corrosion resistant steel bodied connectors which are designed to be assembled to the cable outer conductor using solder shall be gold plated in accordance with MIL-G-45204, type II, class I
3/ These connectors have captivated center contacts
4/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 4
5/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions
6/ Not for use in Army equipment

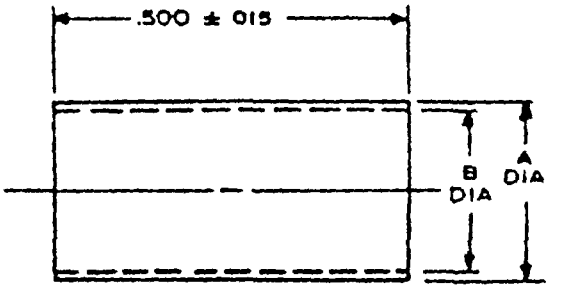
* The latest version of each cable shall be applicable.
★ Cable to be used when performing tests requiring cable except as in note a
a Cable to be used for the +200°C thermal shock tests
∞ Preferred die M22520/5-33 closure B, alternate die M22520/5-03 closure B
Σ Preferred die M22520/5-35 closure B, alternate die M22520/5-03 closure A
& Preferred die M22520/5-41 closure B, alternate die M22520/5-05 closure B, or -09 closure A
\- Preferred die M22520/5-19 closure B, alternate die M22520/5-05 closure A or -11, -57, closure A
ϕ Caution is directed to the application of this cable above 400 MHz Attenuation is tested only at 400 MHz SRL and power handling capabilities are not stipulated herein

NOTE Connectors mate with connectors of the same material only, i e., M39012/59-3001 mates with M39012/55-3001, and M39012/59-4001 mates with M39012/55-4001.

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CENTER CONTACT



CRIMP FERRULE

Dash no.	Contact no 2/	A ± 001	Basic crimp tool 1/	Crimp die or positioner	Crimp tensile min	Color band □	Color band ○
3502 4502	59-10	.041	M22520/1-01	Solder or M22520/1-15	4 lbs	Red	Maroon

Dash no.	Ferrule no 2/	A ± 003	B ± 003	Basic crimp tool 1/	Crimp die or positioner M22520/5-
3502 4502	59-50	250	220	M22520/5-01	-11, -05, -57 closure A and -19 closure B

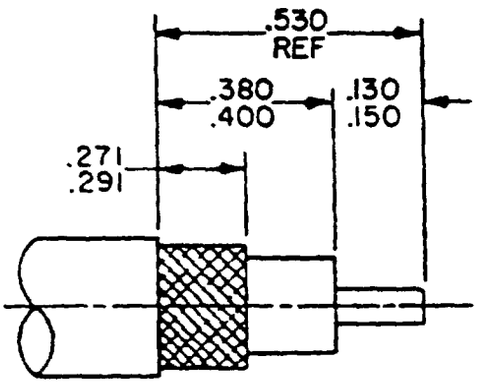
Inches	mm
001	0 03
002	0 05
.003	0.08
005	0 13
010	0.25
015	0 38
024	0.61
025	0 64
.035	0 89
.038	0.97
.041	1 04
.050	1 27
062	1.57
133	3.38
.153	3 89
.193	4.90
220	5 59
.250	6.35
326	8.28
500	12 70

1/ Class 2 tool may be used by OEM (see MIL-C-22520)
2/ Contact numbers and ferrule numbers are for identification only

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Solder or crimp tensile test shall be in accordance with MIL-C-39029.
4. Copyright notice All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner
5. 003 maximum break
6. Color bands shall be positioned so that no coloring material enters the inspection hole.

FIGURE 3 Contact and ferrule dimensions for category D only - Continued

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Inches	mm
130	3 30
150	3 81
271	6.88
291	7 39
380	9 65
400	10 16
530	13 46

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only

FIGURE 4 Cable stripping dimensions for field replaceable connectors

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ENGINEERING DATA

Nominal impedance 50 ohms

Frequency range 0 to 12,400 MHz

Voltage rating The voltage rating shall be in accordance with table II

TABLE II Voltage rating

Cables	Voltage max (at sea level)	Voltage max (70,000 ft)
	V rms	V rms
M17/93-RG178, M17/169-00001	170	45
M17/054-RG122, M17/157-00001, M17/119-RG174, M17/173-00001, M17/113-RG316, M17/172-00001 M17/152-00001	250	65
M17/028-RG058, M17/155-00001, M17/060-RG142, M17/158-00001, M17/084-RG223, M17/167-00001, M17/111-RG303, M17/170-00001, M17/128-RG400, M17/175-00001	335	85

Temperature rating -65°C to +165°C

REQUIREMENTS

Dimensions and configuration: See figure 1

Force to engage and disengage

Longitudinal force Not applicable

Torque 2 inch-pounds, maximum

Coupling proof torque Not applicable

Inspection conditions For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds

Mating characteristics

Reference MIL-STD-348 and figures 2, 4, and 5 for dimensions.

Center contact (socket)

Oversize test pin .0375+ .0001

Test pin finish 16 microinches

Insertion depth .030/ .045

Number of insertions 3

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Insertion force test Steel test pin diameter 0370+ 0001

Insertion depth 050/ 075

Test pin finish 16 microinches

Insertion force 2 pounds, maximum

Withdrawal force test Steel test pin diameter 0355 - .0001

Insertion depth 050/ 075

Withdrawal force 1 ounce, minimum

Test pin finish 16 microinches

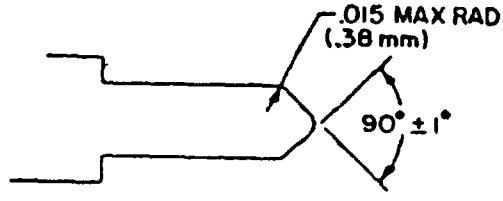


FIGURE 5 Test pin data

Hermetic seal Not applicable

Leakage (pressurized connectors) Connectors shall be mounted in mounting hole shown on figure 1 with mating end capped Test applicable to mounting seal only

Air pressure 30 psi

Duration 30 seconds minimum, 2 minutes maximum

Insulation resistance Method 302 of MIL-STD-202, test condition B 5,000 megohms, minimum

Center contact retention 6 pounds, minimum axial force Applicable to captivated-center-contact connectors only

Radial torque Not applicable

Corrosion (salt spray) Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR)

Cables	VSWR
M17/93-RG178	1 20+0.025 (F) GHz
M17/054-RG122, M17/119-RG174, M17/113-RG316, M17/152-00001	1 15+0 02 (F) GHz
M17/028-RG058, M17/060-RG142, M17/084-RG223, M17/111-RG303, M17/128-RG400	1 15+0 01 (F) GHz

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Swept frequency VSWR test setup

Item 6 VSWR shall be less than 1 025+ 002 F (F in GHz)

Item 16 VSWR shall be less than 1 025+ 002 F (F in GHz)

Second step of VSWR checkout procedure VSWR shall be less than 1 080+ 005 F (F in GHz)

Group B inspection Use step 5, long cable method

Qualification and group C inspection Use step 5, long cable method

Connector durability

Insertion and withdrawal force 500 cycles, minimum at 12 cycles per minute, maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance In milliohms, maximum

	Initial	After environment
Center contact	3 0	4 0
Outer contact	2 0	Not applicable
Braid to body	0 5 1/	Not applicable

Dielectric withstanding voltage at sea level

Method 301 of MIL-STD-202

Cables	V rms
M17/93-RG178, M17/169-00001	500
M17/054-RG122, M17/157-00001, M17/119-RG174, M17/173-00001, M17/113-RG316, M17/172-00001, M17/152-00001	750
M17/028-RG058, M17/155-00001, M17/060-RG142, M17/158-00001, M17/084-RG223, M17/167-00001, M17/111-RG303, M17/170-00001, M17/128-RG400, M17/175-00001	1,000

Vibration, high frequency Method 204 of MIL-STD-202, test condition D No discontinuity permitted

Shock Method 213 of MIL-STD-202, test condition I No discontinuity permitted

Thermal shock Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C High temperature shall be +200°C for connectors using +200°C cables (see tables I and V)

Moisture resistance. Method 106 of MIL-STD-202 No measurements at high humidity Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

1/ Five milliohms are permissible on passivated steel bodied connectors

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Corona level

Altitude - 70,000 feet

Cables	Volts (min)
M17/93-RG178, M17/169-00001	125
M17/054-RG122, M17/157-00001, M17/119-RG174, M17/173-00001, M17/113-RG316, M17/172-00001, M17/152-00001	190
M17/028-RG058, M17/155-00001, M17/060-RG142, M17/158-00001, M17/084-RG223, M17/167-00001, M17/111-RG303, M17/170-00001, M17/128-RG400, M17/175-00001	250

RF high potential withstanding voltage.

Frequency: 5 to 7.5 MHz

Leakage current Not applicable

Cables	Volts (min)
M17/93-RG178, M17/169-00001	335
M17/054-RG122, M17/157-00001, M17/119-RG174, M17/173-00001, M17/113-RG316, M17/172-00001, M17/152-00001	500
M17/028-RG058, M17/155-00001, M17/060-RG142, M17/158-00001, M17/084-RG223, M17/167-00001, M17/111-RG303, M17/170-00001, M17/128-RG400, M17/175-00001	670

Cable retention force. The cable retention force shall be in accordance with table III

TABLE III Cable retention force.

Cable dielectric outer diameter	Pounds (min)	
	Single braid	Double braid
Inches (max)		
036	10	N/A
.067	20	N/A
.110	30	N/A
.122	40	45

Coupling mechanism retention force Not applicable

RF leakage -60 dB minimum tested at a frequency between 2 and 3 GHz

RF insertion loss dB max = .06 x $\sqrt{\text{freq GHz}}$ Test frequency at 6.0 GHz

Part number M39012/59 (dash number from table I or "B" number from table V).

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TABLE IV Group qualification and retention testing

Group	Submission and qualification of any of the following connectors 1/ 2/	Qualifies the following connectors
I	M39012/59- X009	M39012/59- X006 M39012/59- X007 M39012/59- X008 M39012/59- X009 M39012/59- X010
II	M39012/59B X015	M39012/59B X011 M39012/59B X012 M39012/59B X013 M39012/59B X014 M39012/59B X015 M39012/59B X016 M39012/59B X017
III	M39012/59B X022	M39012/59B X018 M39012/59B X019 M39012/59B X020 M39012/59B X021 M39012/59B X022 M39012/59B X023 M39012/59B X024
IV	M39012/59- X028	M39012/59- X025 M39012/59- X026 M39012/59- X027 M39012/59- X028 M39012/59- X029
V	M39012/59- X502	M39012/59- X502

1/ Individual connectors other than listed are self qualifying only

2/ Qualification of connectors qualifies connectors of the same body material and finish only

X denotes material

For qualification retention, where more than one part is listed in a group in this column, data may be supplied on any of those parts in order to retain qualification for those parts in the corresponding right hand column. The part does not necessarily have to be the part initially qualified. NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers by qualifying the one connector It is not necessary that such connectors be in the same group Each connector, however, must be marked with its own appropriate part number For group qualification, the connectors must be of similar design

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TABLE V Category B - Nonfield replaceable (special tools may be required).

Not for Air force or Navy use For OEM use only

M39012/59B	Applicable cable number *	Dimensions	Inches-millimeters maximum 1/ 2/
3011 3/ 4/ 4011 3/ 4/	M17/93-RG178* M17/169-00001φ		
3012 3/ 4/ 4012 3/ 4/	M17/119-RG174 M17/173-00001φ M17/113-RG316* M17/172-00001φ		
3013 3/ 4/ 4013 3/ 4/	M17/54-RG122* M17/157-00001φ		
3014 3/ 4/ 4014 3/ 4/	M17/28-RG058* M17/155-00001φ	A	1.265 (32.13)
3015 3/ 4/ 4015 3/ 4/	M17/60-RG142*@ M17/158-00001φ		
3016 3/ 4/ 4016 3/ 4/	M17/84-RG223* M17/167-00001φ		
3017 3/ 4/ 4017 3/ 4/	M17/111-RG303* M17/170-00001φ		
3018 5/ 4018 5/	M17/93-RG178* M17/169-00001φ		
3019 5/ 4019 5/	M17/119-RG174 M17/173-00001φ M17/113-RG316* M17/172-00001φ		
3020 5/ 4020 5/	M17/54-RG122* M17/157-00001φ		
3021 5/ 4021 5/	M17/28-RG058* M17/155-00001φ		

See footnotes at end of table

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TABLE V Category B - Nonfield replaceable (special tools may be required) - Continued

Not for Air force or Navy use For OEM use only

M39012/598	Applicable cable number «	Dimensions	Inches-millimeters maximum 1/ 2/
3022 5/	M17/60-RG142*@	A	1.265(32.13)
4022 5/	M17/158-00001ϕ		
3023 5/	M17/84-RG223*		
4023 5/	M17/167-00001ϕ		
3024 5/	M17/111-RG303*		
4024 5/	M17/170-00001ϕ		

1/ Millimeters are in parentheses

- All corrosion resistant steel bodied connectors which are designed to be assembled to the cable outer conductor using solder shall be gold plated in accordance with MIL-G-45204, type II, class I

3/ These connectors will be furnished with noncaptive center contacts.

4/ Inactive for new design

5/ These connectors have captivated center contacts

« The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in note Σ

@ Cable to be used for the +200°C thermal shock tests.

ϕ Caution is directed to the application of this cable above 400 MHz Attenuation is tested only at 400 MHz SRL and power handling capabilities are not stipulated herein

NOTES

1 Noncaptive-center-contact connectors shall not be used on Air Force equipment

2 Connectors mate with connectors of the same material only; Example: M39012/59-3001 mates with M39012/55-3001, and M39012/59-4001 mates with M39012/55-4001

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TABLE VI Maintenance replacements for category B

Category B number*	Category C dash number	Category D dash number	Category D dash number
BX011	X025	X006	---
BX012	X026	X007	---
BX013	X027	X008	---
BX014	X029	X009	---
BX015	X028	X009	X502
BX016	X028	X009	---
BX017	X029	X010	---
BX018	X025	X006	---
BX019	X026	X007	---
BX020	X027	X008	---
BX021	X029	X009	---
BX022	X028	X009	X502
BX023	X028	X009	---
BX024	X029	X010	---

* Category B connectors are for original installation only They will not be stocked or acquired by the Government

X denotes material

The material of the item shall be the same material as the item being replaced Example 55-3011 (corrosion resistant steel) replaces 55-3025.

TABLE VII Cross reference of part numbers

Part number M39012/59B 1/	Superseded part number M39012/59-
X011	X011
X012	X012
X013	X013
X014	X014
X015	X015
X016	X016
X017	X017
X018	X018
X019	X019
X020	X020
X021	X021
X022	X022
X023	X023
X024	X024

1/ The 'B' part number is required marking for connectors manufactured after 3 April 1987 The connectors that are in stock or distribution that were previously qualified and marked with the old part number shall also be considered acceptable for Government use until stock is purged (Applies to category 'B' part number change on M39012/XX8XXXX)

The material of the item shall be the same material as the item being replaced Example 55B3011 (corrosion resistant steel) replaces 55-3025

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Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities
Army - EA, MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AT, AV
Navy - AS, MC, OS
Air Force - 19

Preparing activity.
Army - CR

Agent·
DLA - ES

(Project 5935-3784)

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